

# Gavin J. Qu

[gavinqu.dev@gmail.com](mailto:gavinqu.dev@gmail.com) | (425)-324-7177 | [GitHub](#) | [LinkedIn](#) | [Blog](#) | U.S. Citizen

## Education

**University of Essex** | November 2024 **MSc in Economics**

**Honors:** Distinction (~4.0 GPA)

**Relevant coursework:** Public Policy Evaluation (Causal Inference), Data Science Economics, Graduate Econometric Methods, Mathematical Methods (Optimization & Dynamic Programming), Advanced Microeconomic Theory, Behavioral Economics, Advanced Macroeconomic Theory.

**University of Washington** | June 2022 **BA in Economic & BA in Philosophy**

**Honors:** Dean's List (2020-2022)

**Relevant coursework:** Calculus I-III, Differential Equations, Statistics, Market Design, Econometrics, Public Economics, Environmental Economics, Intermediate Micro/Macro, Industrial Organization.

**UIUC** | 2025-2027 **Math Preparation**

*MATH 416: Abstract Linear Algebra; MATH 314: Logic and Proof; MATH 447 Real Analysis.*

**Professional Certificates:** Data Structures & Algorithms; Object-Oriented Programming in Java; CS50.

## Research and Projects

**MSc Thesis | University of Essex | Advisor: David Zentler-Munro**

"From Medical Charts to Market Forces: How Health Types Shape Labour Market Outcomes" ([GitHub Link](#))

- Applied unsupervised machine learning techniques to the UKHLS (Understanding Society) Panel Data (2009–2022) using Python.
- Identified health-type heterogeneity using k-means clustering and modeled its specific effects on employment, earnings, and mortality dynamics.
- Utilized econometric modeling to analyze health-based disparities in the labor market.
- Implemented data science techniques to handle large-scale longitudinal data, ensuring results were statistically robust.

## Research Communication & Open-Source

- **Creator, [econ-phd-advice](#) (open-source guide):** Built and maintain a community-driven Quarto/GitHub resource hub for economics PhD and predoc applicants.
- **Substack & Blog on Economics and Statistics.**

## Experience

**Pre-doctoral and PhD preparation**

*May 2026 – Present*

**Costco Travel**

*May 2025 – May 2026*

- Conducted research-style exploratory analysis to identify drivers of operational variance; produced clear summaries for decision-making.
- Implemented statistical/econometric models (e.g., logistic regression) to quantify relationships and generate interpretable effect-based diagnostics.
- Applied machine learning methods (Random Forest, Gradient Boosting, NLP) to improve prediction and compare performance against baseline statistical models.
- Engineered reproducible Python/SQL pipelines to clean, merge, and validate large operational datasets (schema checks, missingness, outliers), improving data integrity and reducing manual work.
- Developed binary-classification models for price-variance detection (92% F1 score) with cross-validation.
- Standardized documentation and versioned workflows (LaTeX, Git) and automated ETL with Python/Bash to support scale and workflow reproducibility.

- **Technical Leadership:** Designed and hosted departmental workshops on SQL, statistical thinking, and causal inference.

**Costco Corporate**, Issaquah, WA

2022 — 2023

- Conducted forecasting and EDA to identify operational inefficiencies.

**Math Instructor | Juni Learning**, Seattle, WA

2021 — 2022

- Taught mathematical problem-solving and analytical reasoning from Pre-Algebra through AP Calculus.
- Assisted in designing curriculum on Economics and Personal Finance.

## Technical Skills & Research Interests

---

**Field of Interests:** Labor, Health, Behavioral, Economics of Innovation.

**Methodological Interests:** Causal Inference, Bayesian, and Machine Learning Methods.

**Causal Inference & Econometrics:** Difference-in-Differences (DiD), Regression Discontinuity Design (RDD), Instrumental Variables (2SLS/LATE), Propensity Score Matching, Randomized Controlled Trials (RCTs), Panel Data Analysis, Quantile Regression, Causal Forest, Double-debiased ML.

**Machine Learning:** Regularization (Lasso/Ridge), Ensemble Methods (Random Forest, Gradient Boosting), Dimensionality Reduction (PCA/PCR), Generalized Additive Models (GAMs), Classification (LDA/QDA), Neural Networks (CNN/RNN), Cross-Validation & Bootstrapping.

**Mathematical Methods:** Dynamic Programming, Constrained & Unconstrained Optimization, Differential & Difference Equations, Maximum Principle.

**Programming & Tools** – Python, R, Julia, SQL, Java, Git, *LaTeX*, Bash, Tableau, Claude Code.